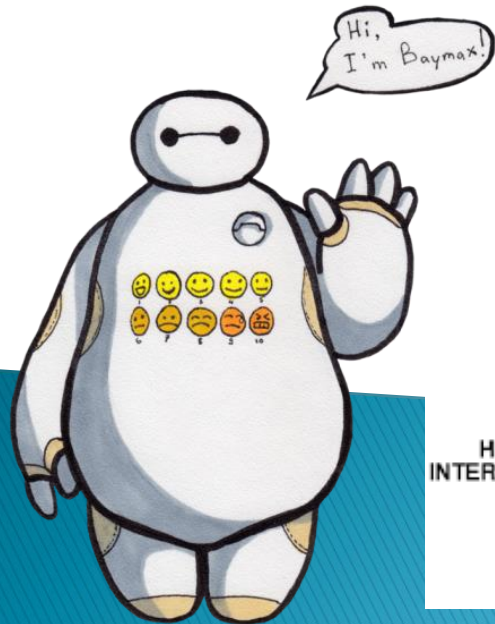


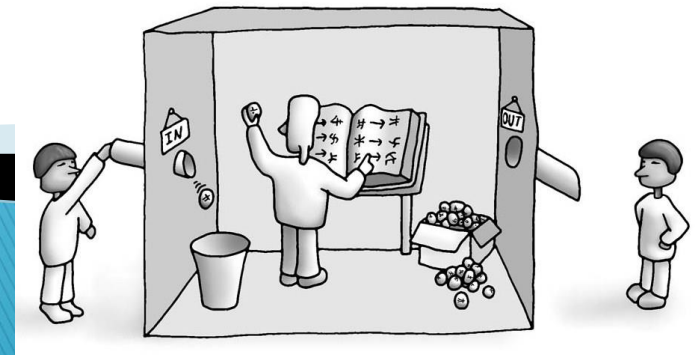
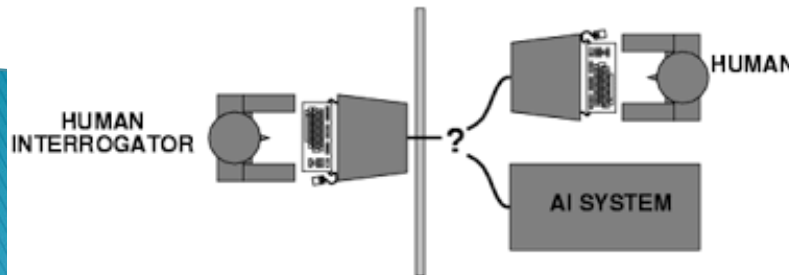
Please record your attendance today at **SEAtS**

COMP1008 - Fundamentals of AI

Course Introduction

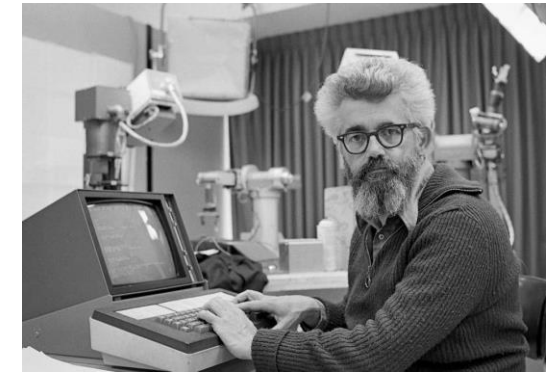


Professor Rong Qu
rong.qu@nottingham.ac.uk



What is AI? – Definitions

- ▶ Coined in 1956: "the science and engineering of making **intelligent** machines"



John McCarthy, 1956

- ▶ A branch of computer science and engineering that deals with **intelligent behaviour**, **learning**, and **adaptation** in machines.

Wikipedia, 2009



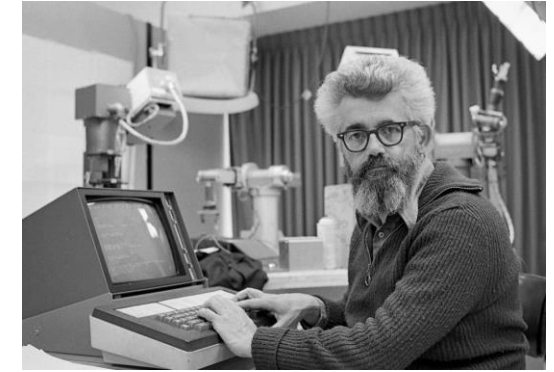
- ▶ Create a computer mind that can **think** like a human.

BBC: What is Artificial Intelligence? 2015

- ▶ ... capability of computational systems to perform tasks typically associated with human intelligence, such as **learning**, **reasoning**, **problem-solving**, **perception**, and **decision-making**

Wikipedia, 2026

What is AI? – Definitions



	Weak AI	Strong AI
Scope	Single specific tasks	Broad, general purpose
Self-awareness	No	Yes (theoretically)
Learning ability	Limited on specific tasks	General human like
Status	Exists, widely used	Theoretical
Examples	Siri, AlphaZero	Human-like robots / systems

Weak vs. Strong AI (AGI – Artificial General Intelligence)

CPMP1008: An Overview of This Session

- ▶ A brief history of AI
 - AI Milestones
 - Themes: top-down vs. bottom-up
- ▶ AI fundamental issues
 - Two AI experiments: Turing test, Chinese room experiment
 - One(?) definition: AI
- ▶ Course content

Milestones in AI

- ▶ 1943 WW2
 - **Turing Test**: influenced AI, till now
- ▶ 1956 Birth of AI
 - **AI term** coined, Dartmouth College
 - Early AI: Symbolic AI, Search techniques, Nature Language Processing, etc.)
 - **Top-down** approach (deductive): pre-processed to **do** things
vs.
 - **Bottom-up** approach (inductive): simulate brains, **learn** new behaviours and knowledge



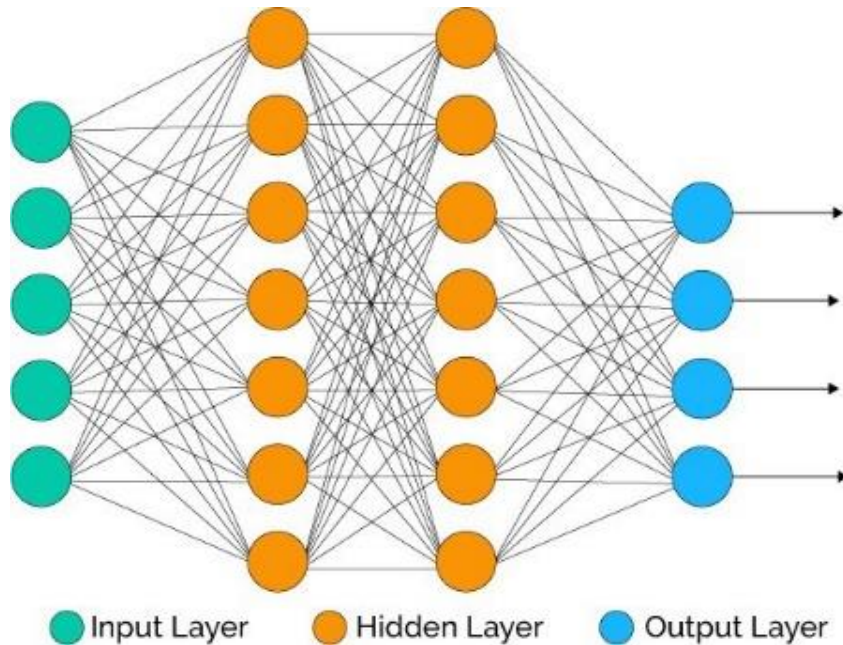
Milestones in AI - Examples

- ▶ Machine translations: NRC, 1950s', translation from Russian to English
 - “*The spirit is willing but the flesh is weak*” produced
 - “*The vodka is good but the meat is rotten*”
 - Requires **knowledge** to establish context
 - Literal translation vs. free translation: **understanding**
 - Are these AI systems (refer to the definitions of AI)?
- **Top-down** approach: pre-processed to **do** things
vs.
- **Bottom-up** approach: simulate brains, **learn** new behaviours and knowledge



Milestones in AI – AI winter

- ▶ 1970s: imitate biology
 - **AI winter:** Neurons (in ANN) not able to implement basic functions



AI	COMP1008	CS Course
Bottom-up	Machine learning	COMP3009



Milestones in AI (continued)

- ▶ 1997 **Deep Blue**
 - 11 May 1997, AI Game milestone
- ▶ 2011 **IBM Watson**
 - Won US quiz show Jeopardy
- ▶ 2014 **Turing test** “passed”
 - Chatbot Eugene Goostman
- ▶ 2016 **AlphaGo**
 - All open board games conquered
- ▶ 2022 **ChatGPT**
 - Strong AI (AGI)?
 - Recent **LLMs** (GPTs, Gemini, Claude, LLaMA, DeepSeek, etc.)



Milestones in AI (continued)

- ▶ 1997 **Deep Blue**
 - 11 May 1997, AI Game milestone
- ▶ 2011 **IBM Watson**
 - Won US quiz show Jeopardy
- ▶ 2014 **Turing test** “passed”
 - Chatbot Eugene Goostman
- ▶ 2016 **AlphaGo**
 - All open board games conquered
- ▶ 2022 **ChatGPT**
 - Strong AI (AGI)?

AI	COMP1008	CS Courses
Top-down	Knowledge Representation and Reasoning (KRR)	COMP3008
Mixed	NLP + Machine learning + KRR + Search	COMP1008 COMP3009
Bottom-up	Natural language processing (NLP), KRR	COMP1008 COMP3074
Mixed	Alpha-beta Search + Deep learning	COMP1008 COMP3009
Mixed	Deep learning + NLP	COMP3009 COMP3074

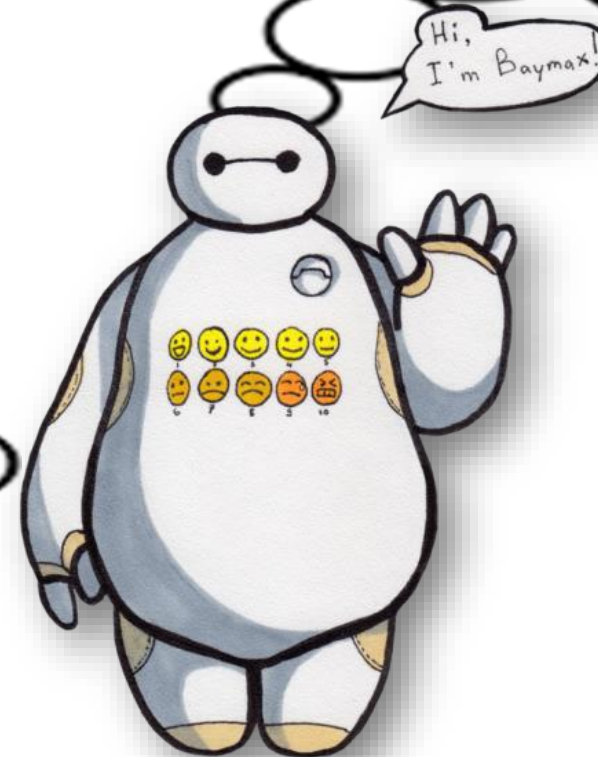
COMP1008 AI techniques vs. AI courses at the School

AI in the school

COMP1008 - Fundamentals of AI
COMP3003 - Individual Project

COMP2001 - AI Methods
COMP2005 - Image Processing
COMP3007 - Computer Vision
COMP3004 - Design Intelligent Agents
COMP3008 - Symbolic AI
COMP3009 - Machine Learning
COMP3074 - Human AI Interaction

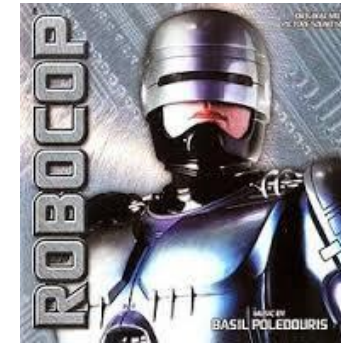
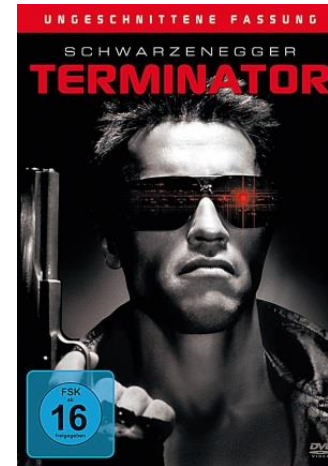
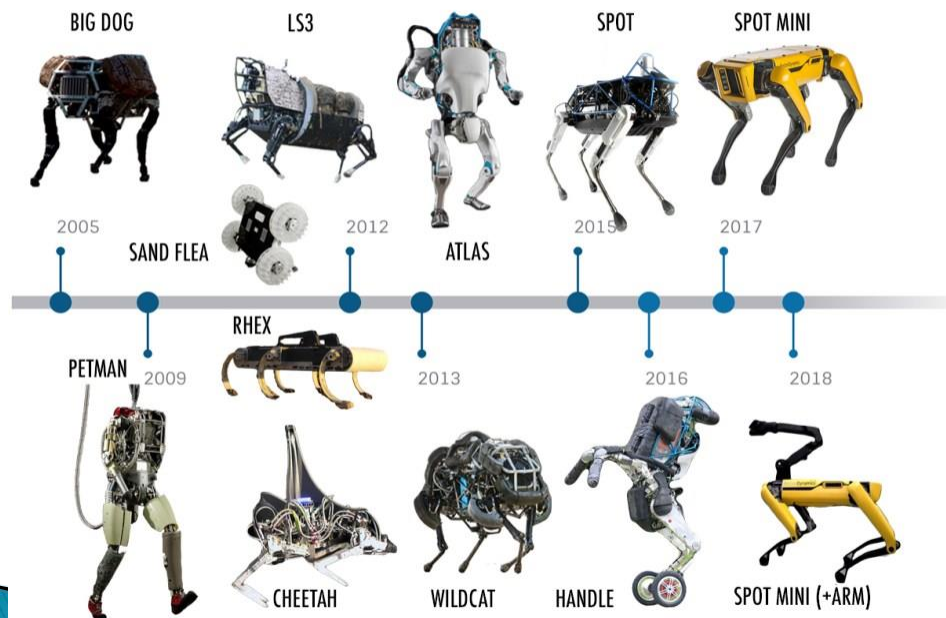
COMP4030 - Data Science and Machine Learning
COMP4033 - Fuzzy Sets and Fuzzy Systems
COMP4041 - Linear and Discrete Optimisation
COMP4124 - Big Data Learning and Tech
COMP4034 - Robotic Systems



What is AI?

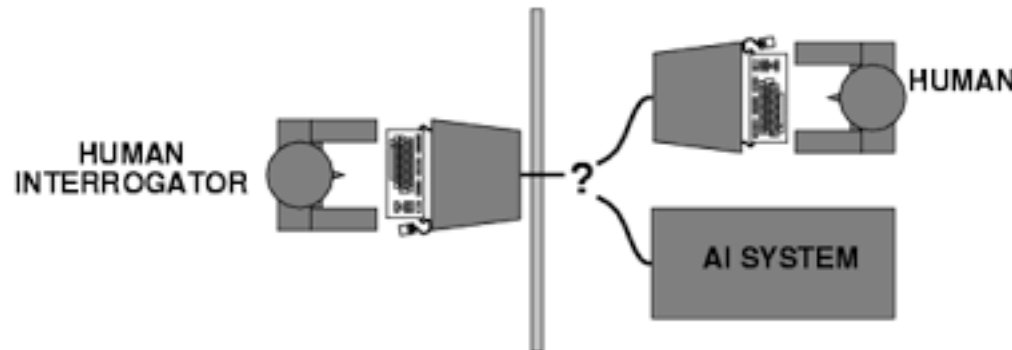
- ▶ Robots in movies vs. in real-life
- ▶ Can machines ever be **intelligent**?

BOSTON DYNAMICS

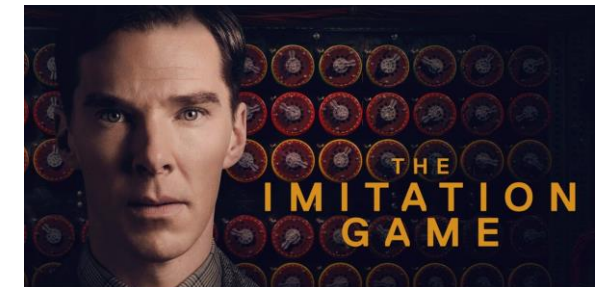


The Turing Test - Definition

- ▶ Turing: father of Computer Science, mathematician, philosopher, code breaker
 - 1936-38: Princeton University, PhD in logic, algebra, number theory
 - The Turing machine: *On Computable Numbers*
 - 1939-40: Devises Bombe, machine for *Enigma* decryption
 - 1950: Philosophical paper on *machine intelligence*: *The Turing Test*
- ▶ Task: which candidate is the machine, *only by asking them questions*
- ▶ If the machine can fool the interrogator *30% of the time*, the machine is considered **intelligent**



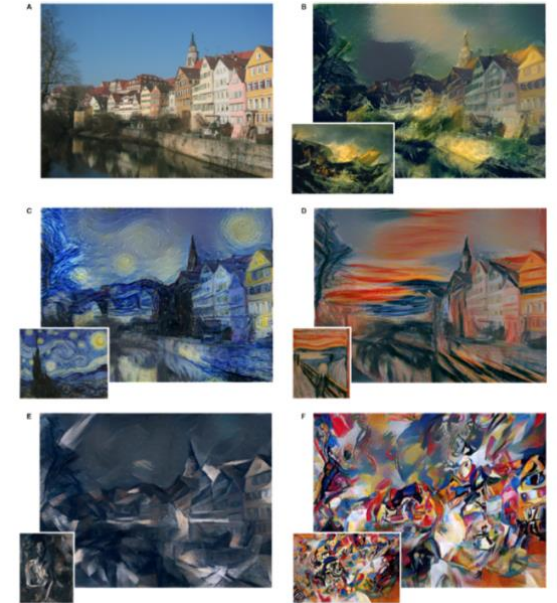
Question: "What is 35,076 divided by 4,567?"
Answer A: ???? Answer B: 7.6803153



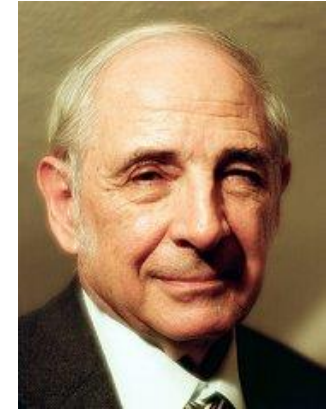
The Turing Test - Examples

- ▶ If the Turing Test was passed, Turing would conclude that the machine was **intelligent**
 - Suggested a way of saying if machines **is** intelligent, or at least **act** intelligently?
 - A satisfactory **operational** definition of intelligence

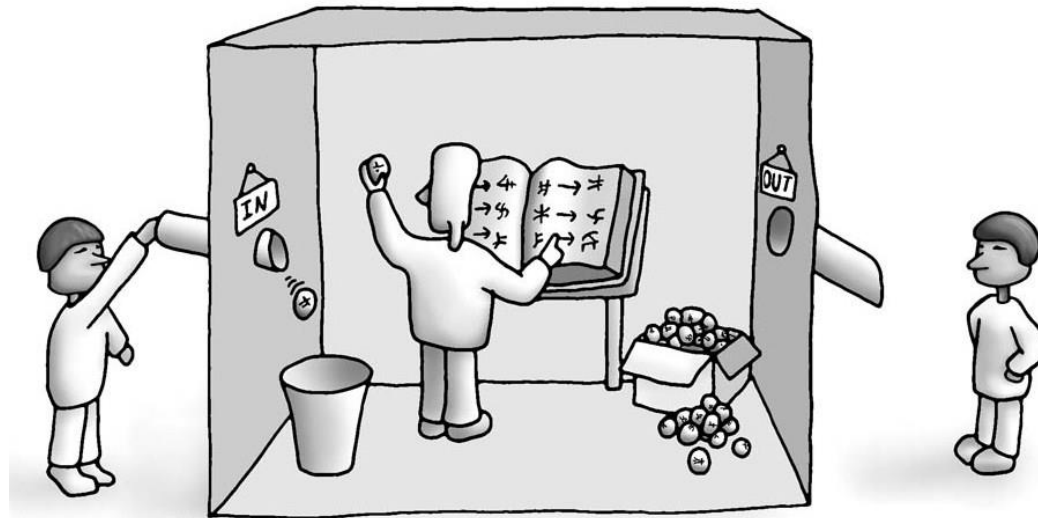
- ▶ Real life Turing test examples
 - Online/AI games
 - Chatroom, Content Generator
 - AI Art generators: music / painting / poetry composing programs



The Chinese Room Experiment



- ▶ A thought experiment by **John Searle***
 - Simple rule processing system: a rule book
 - The “rule processor”: intelligent but has **no understanding** of Chinese
- ▶ The system does **act** as it understands Chinese!
 - ▶ **Behaving** intelligently: not enough to prove a computer **is** intelligent



* Searle, J.R. 1980. Minds, brains and programs. Behavioural and Brain Sciences, 3: 417-457, 1980

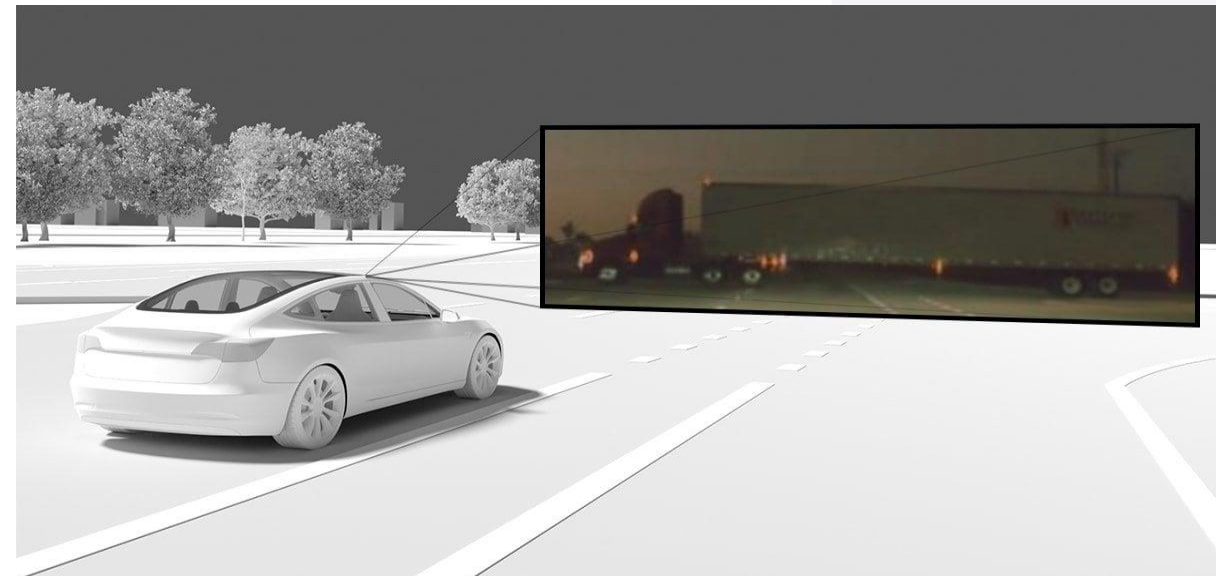
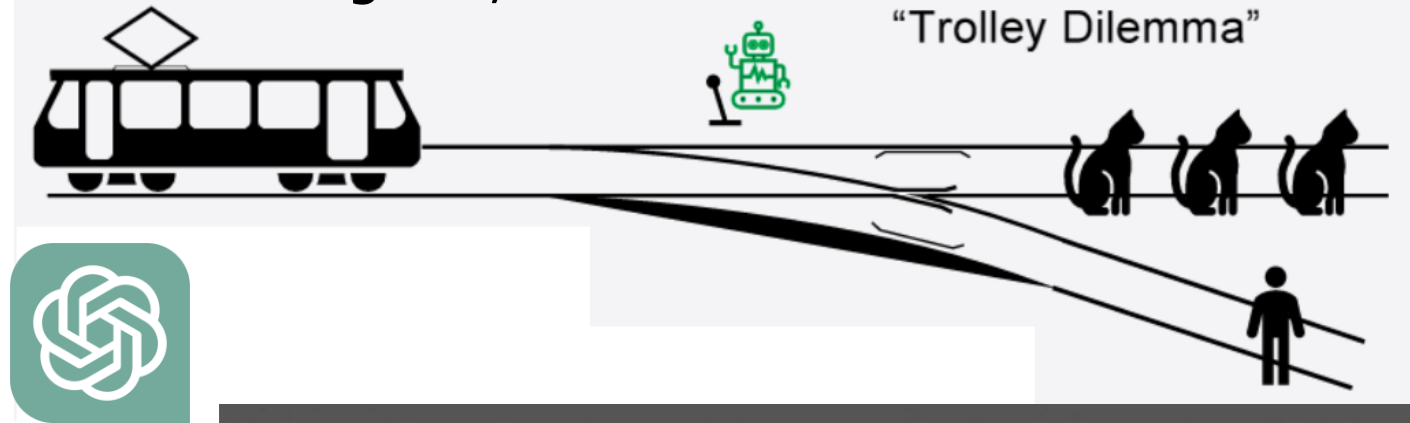
Predictions in AI

- ▶ [Herbert Simon](#), 1957
 - “Within 10 years a computer will be a chess champion”
- ▶ [Alan Turing](#)
 - “In some stage we should have to expect the machines to take control”
- ▶ [Ray Kurzweil](#):
 - *The Age of Intelligent Machines*, 1990: “A computer ... beating a chess champion by 1998”
 - *The Singularity is near*, 2005 book: Superintelligence possesses intelligence far surpassing that of the brightest human minds
 - 2014: “Robots will be smarter than us all by 2029”
- ▶ Your prediction?!
 - Can machines ever be [intelligent](#)?
 - If so, when?



Responsible & Ethical AI

- ▶ Traditional ethics was built for **human moral** agents, and for **human decisions**
- ▶ AI breaks these assumptions
- ▶ **Privacy** and security of data
- ▶ **Bias** and fairness
- ▶ **Transparency** and explainability
- ▶ **Accountability** and responsibility
- ▶ **Safety**, etc.



COMP1008	CS Course
Image Processing	COMP2005
Human AI Interaction	COMP3074
Professional Ethics	COMP3020
Machine Learning	COMP3009

Responsible & Ethical AI

- ▶ Traditional ethics was built for human moral
- ▶ AI breaks these assumptions
- ▶ Privacy and security of data
- ▶ Bias and fairness
- ▶ Transparency and explainability
- ▶ Accountability and responsibility
- ▶ Safety, etc.



Fake Obama created using AI tool to make phoney speeches

Course Convenors

- ▶ Professor Rong Qu
 - PhD in Computer Science, University of Nottingham
 - Evolutionary Computation, Optimisation, Machine learning
- ▶ Dr Luis Figueredo
 - PhD in Robotics, University of Brasilia
 - Robot Manipulator Control, Multiple Robotic Arms, Safe and Reliable human-robot Manipulation
- ▶ Computing session TAs
 - Ifeoma Ogbodo, Warren Jackson, Bogna Liziniewicz, Yuhe Gong, Yu Zhao



Course Info

- ▶ **Lectures:** 18hrs, Mondays 4pm-6pm, Business School South Auditorium B52
- ▶ **Computing sessions:** 14hrs incl. preparation, Thursdays 11.00 or 12.00, CS-A32
 - Handout materials of exercises: release on Tuesdays
 - Jupyter Notebook: data analysis / visualisation, machine learning, example projects/demos
- ▶ **Coursework:** ~25hrs
- ▶ **Exam revision, self-study:** ~38 hrs
- ▶ **Office hours:** ~5hrs
- ▶ **Moodle:** all slides, recordings, materials, assessments info, Q&As
- ▶ **Emails:** we aim to answer questions within two working days

Course Info – Assessment

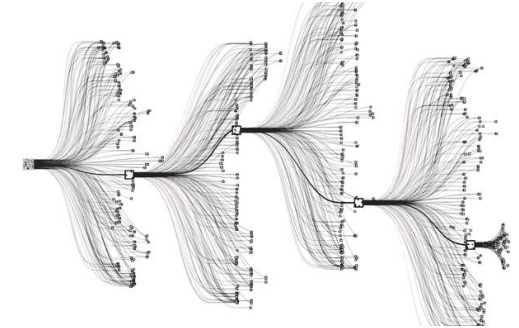
- ▶ 75% examination **75 marks**
 - 2hrs questions, in-person ExamSys
 - Covers all lecture content, unless specified otherwise (e.g. excl. computing)
- ▶ 25% coursework **25 marks**
 - Questions on machine learning applied to classification / regression tasks
 - Skills required are supported in computing sessions
 - To be released in Feb, TBC*
 - Submission deadline late March, TBC*

TBC*: please check Moodle regularly

Course Info – Lecture Schedule

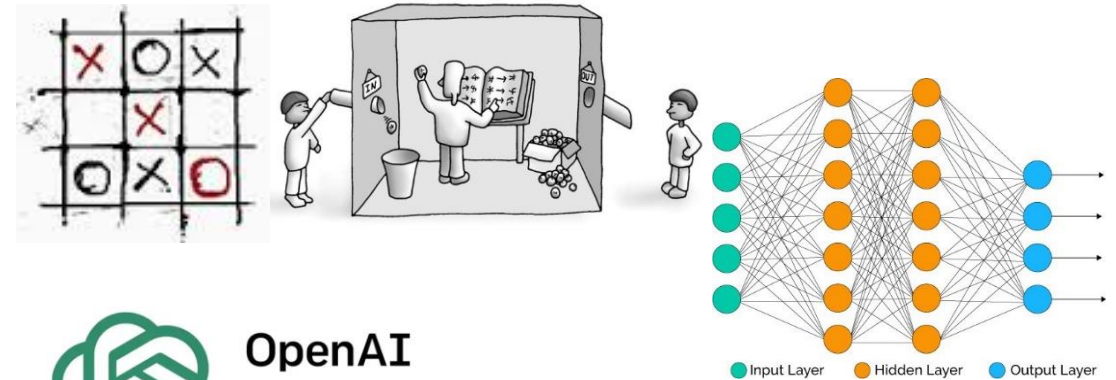
► Part I: Fundamental Issues

- AI milestones, AI experiments, brief history, developments



► Part II: Machine learning

- Artificial Neural Networks (ANN)
- Other techniques (Regression, Decision Tree)



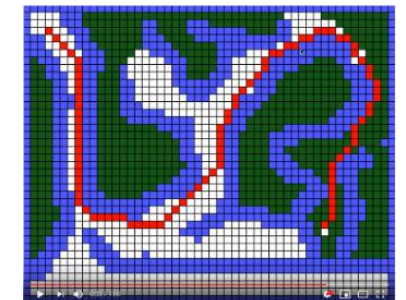
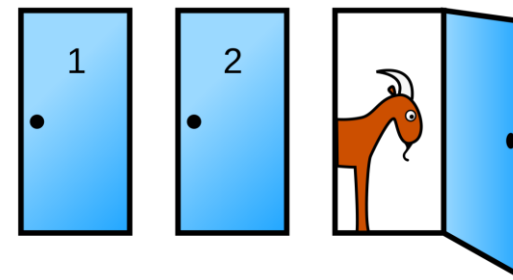
► Part III: Intelligent techniques

- Bayes Rule, KRR
- Natural Language Processing, LLMs



► Part IV: Tree search techniques

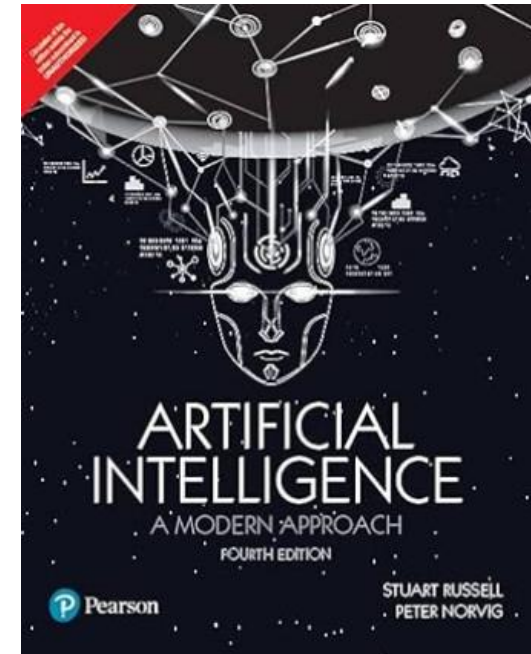
- Combinatorial explosion
- Depth First, Breadth First, A*
- Game Playing (MiniMax, Alpha-beta pruning)



Course Info – Textbooks (lectures)

► Artificial Intelligence – A Modern Approach (Russell / Norvig), 1995 - 2021

- Most used comprehensive textbook in AI, <https://aima.cs.berkeley.edu/contents.html>
- Reference book: You don't have to learn the **whole** book
- Chap 1.1, 1.2: Introduction, Philosophy
- Chap 3: Solving Problems by Search
- Chap 3.5: Informed (Heuristic) Search
- Chap 5.3: Alpha-beta Tree (Game playing)
- Chap 7.4: Knowledge Representation Reasoning
- Chap 12.5: Bayes' Rule
- Chap 19.2: Supervised Learning
- Chap 19.3: Decision Tree
- Chap 19.6: Regression
- Chap 23: Natural Language Processing



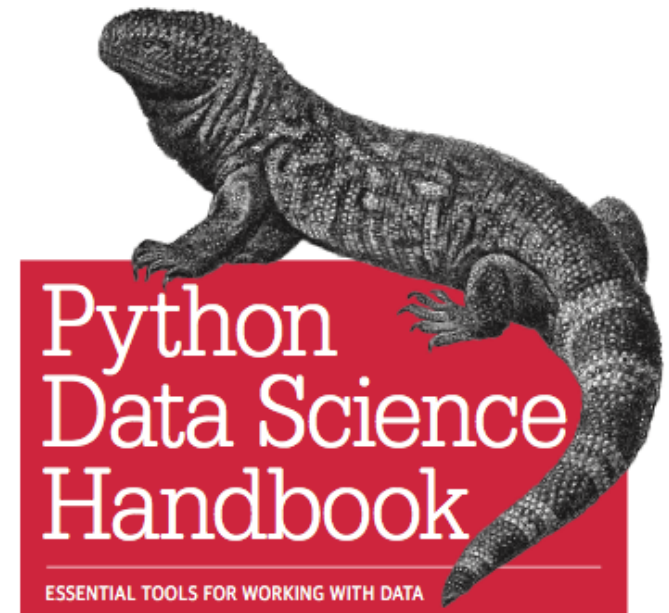
Course Info – Textbooks (computing)

▶ Python Data Science Handbook (Jake VanderPlas)

- Examples of machine learning using [Jupyter Notebook](#)
- A lot of [free online resources](#)
- Reference book: You don't have to learn the **whole** book

 DataSpell DataSpell is an IDE for exploratory data analysis and prototyping machine learning models. It combines the interactivity of Jupyter notebooks with the intelligent Python and R coding assistance of PyCharm in one user-friendly environment. Install	 CMD.exe Prompt 0.1.1 Run a cmd.exe terminal with your current environment from Navigator activated Launch	 JupyterLab 3.3.2 An extensible environment for interactive and reproducible computing, based on the Jupyter Notebook and Architecture. Launch	 Jupyter Notebook 6.4.0 Web-based, interactive computing notebook environment. Edit and run human-readable docs while describing the data analysis. Launch	 Powershell Prompt 0.0.1 Run a Powershell terminal with your current environment from Navigator activated Launch	 Qt Console 5.3.0 PyQt GUI that supports inline figures, proper multiline editing with syntax highlighting, graphical calltips, and more. Launch
 Spyder 5.1.5 Scientific Python Development Environment. Powerful Python IDE with advanced editing, interactive testing, debugging and introspection features Launch	 Datalore Kick-start your data science projects in seconds in a pre-configured environment. Enjoy coding assistance for Python, SQL, and R in Jupyter notebooks and benefit from no-code automations. Use Datalore online for free. Launch	 Deepnote Deepnote is a notebook built for collaboration. Create notebooks in your browser, spin up your conda environment in seconds and share with a link. Launch	 IBM Watson Studio Cloud IBM Watson Studio Cloud provides you the tools to analyze and visualize data, to cleanse and shape data, to create and train machine learning models. Prepare data and build models, using open source data science tools or visual modeling. Launch	 ORACLE Cloud Infrastructure Oracle Data Science Service OCI Data Science offers a machine learning platform to build, train, manage, and deploy your machine learning models on the cloud with your favorite open-source tools Launch	 PyCharm Professional A Full-Featured IDE by JetBrains for both Scientific and Web Python development. Supports HTML, JS, and SQL. Install

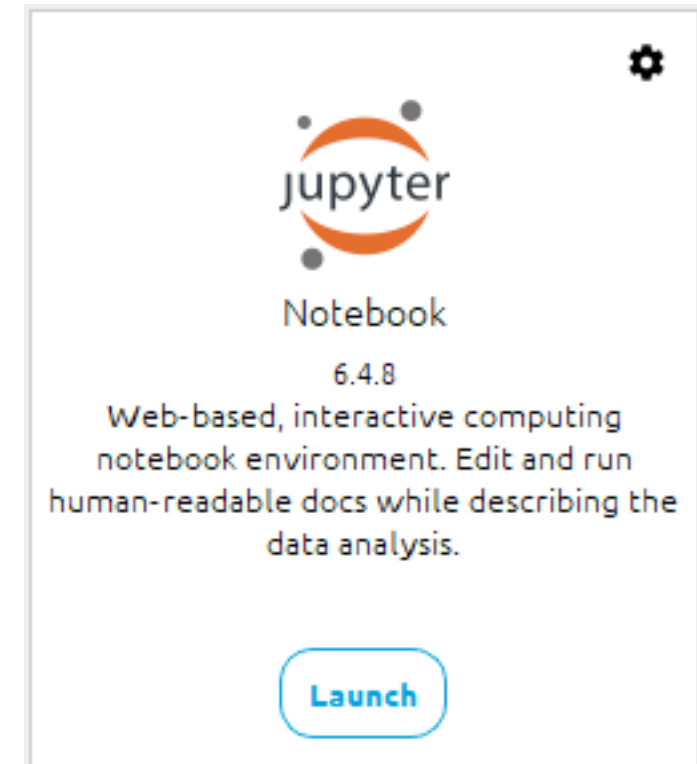
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Jake VanderPlas

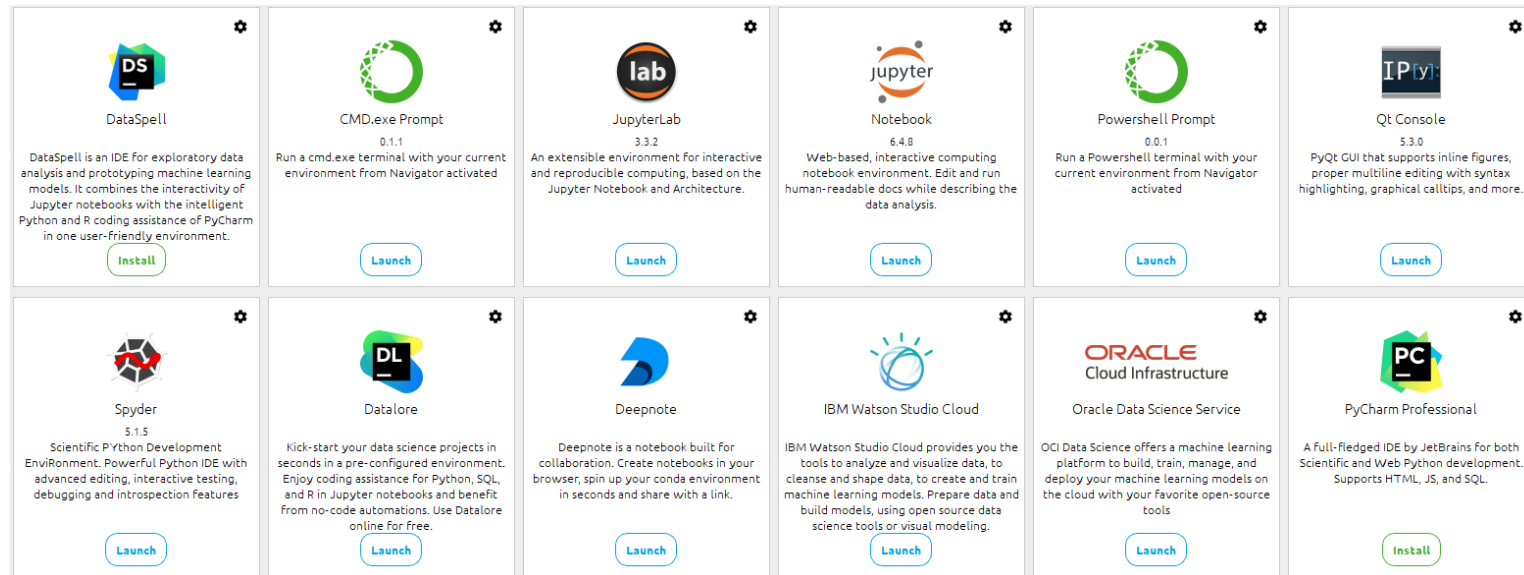
Computing Exercises

- ▶ **COMP1008 Computing Sessions** start from Thursday 29 Jan 2026
 - Exercises released on Mondays. Please go through the materials before the computing sessions, and come to A32 if you have any problems
 - Not assessed, but provide [support to coursework](#) and [data analysis machine learning skills](#) to complete the coursework
- ▶ **How do I get help?**
 - In-person: [timetabled computing sessions](#)
 - Thursdays 11am-12pm, 12pm-1pm, A32 (repeated sessions, please check your group in Moodle)
 - Emails: [Questions to convenors, TAs](#)
 - We aim to respond within two working days in term time



Computing Exercises

- ▶ Anaconda Navigator / Jupyter Notebook
 - Web based interactive UI
 - **Exercise 0: Jupyter Notebook - Basics**



- **Anaconda:** A32 desktop; Local installation on your laptop; Virtual desktop